

BASICS OF ELECTRICAL ENGINEERING

(EE113ATD)

Unit-III

Magnetic Circuits:

Analogy between electric and magnetic circuits, self-inductance and mutual inductance, coefficient of coupling.

DC Machines:

DC Generator: Construction and principle of operation, induced emf expression, classification of generators with relation between induced emf and terminal voltage.

DC Motor:

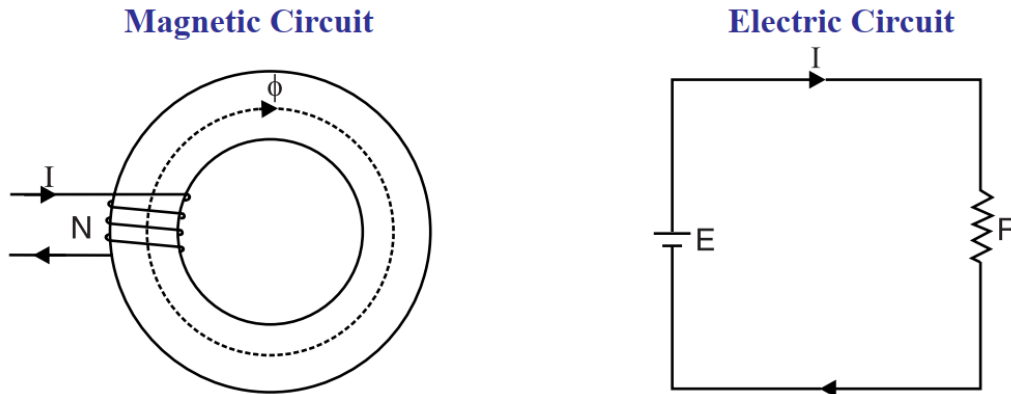
Back emf and its significance, Torque equation, classification, characteristics and applications.

EV Motors: Introduction to BLDC, PMSM – Construction and principle of operation.

Magnetic Circuits:

Analogy between electric and magnetic circuits

There are many points of similarity between magnetic and electric circuits. However, the two circuits are not analogous in all respects. A comparison of the two circuits is given below in the tabular form.



Similarities

1. The closed path for magnetic flux is called a magnetic circuit.	1. The closed path for electric current is called an electric circuit.
2. Flux, $\phi = \frac{\text{m.m.f.}}{\text{reluctance}}$	2. Current, $I = \frac{\text{e.m.f.}}{\text{resistance}}$
3. m.m.f. (ampere-turns)	3. e.m.f. (volts)
4. Reluctance, $S = \frac{l}{a\mu_0\mu_r}$	4. Resistance, $R = \rho \frac{l}{a}$
5. Flux density, $B = \frac{\phi}{a} \text{ Wb/m}^2$	5. Current density, $J = \frac{I}{a} \text{ A/m}^2$
6. m.m.f. drop = ϕS	6. Voltage drop = IR
7. Magnetic intensity, $H = NI/l$	7. Electric intensity, $E = V/d$
8. Permeance	8. Conductance.
9. Permeability	9. Conductivity

Dissimilarities

1. Truly speaking, magnetic flux does not flow.	1. The electric current actually flows in an electric circuit.
2. There is no magnetic insulator. For example, flux can be set up even in air (the best known magnetic insulator) with reasonable m.m.f.	2. There are a number of electric insulators. For instance, air is a very good insulator and current cannot pass through it.

3. The value of μ_r is not constant for a given magnetic material. It varies considerably with flux density (B) in the material. This implies that reluctance of a magnetic circuit is not constant rather it depends upon B. 4. No energy is expended in a magnetic circuit. In other words, energy is required in creating the flux, and not in maintaining it.	3. The value of resistivity (ρ) varies very slightly with temperature. Therefore, the resistance of an electric circuit is practically constant. This salient feature calls for different approach to the solution of magnetic and electric circuits. 4. When current flows through an electric circuit, energy is expended so long as the current flows. The expended energy is dissipated in the form of heat.
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Calculation of Ampere-Turns

In any magnetic circuit, flux produced is given by ;

$$\text{Flux, } \phi = \frac{\text{m.m.f.}}{\text{reluctance}} = \frac{AT}{(l/a\mu_0\mu_r)}$$

$$\begin{aligned} \therefore AT \text{ required} &= \phi \times \frac{l}{a\mu_0\mu_r} = \frac{\phi}{a} \times \frac{l}{\mu_0\mu_r} \\ &= \frac{B}{\mu_0\mu_r} \times l && \left(\because B = \frac{\phi}{a} \right) \\ &= H \times l && (\because H = B/\mu_0\mu_r) \end{aligned}$$

i.e. **AT required for any part = Field strength H in that part × length of that part of magnetic circuit**

Inductance

Inductance is the tendency of an electrical conductor to oppose a change in the electric current flowing through it. L is used to represent the inductance, and Henry is the SI unit of inductance. 1 Henry is defined as the amount of inductance required to produce an emf of 1 volt in a conductor when the current change in the conductor is at the rate of 1 Ampere per second.

An electric current flowing through a conductor creates a magnetic field around it. The strength of the field depends upon the magnitude of the current. The generated magnetic field follows any changes in the current, and from Faraday's law of induction, we know that changing the magnetic field induces an electromotive force in the conductor. Considering this principle, inductance is defined as the ratio of the induced voltage to the rate of change of current causing it. The electronic component designed to add inductance to a circuit is an inductor.

Factors Affecting Inductance

Following factors affect the inductance in a circuit:

Number of Wire Turns in the Coil

Inductance is greater when the number of turns of wire in the coil is greater. More coils of wires indicate a greater amount of magnetic field force for a given amount of coil current.

Coil Area

Inductance is proportional to the coil area. Greater the coil area, the greater the inductance. Greater coil area presents less opposition to the formation of magnetic field flux for a given amount of field force

Core Material

The greater the magnetic permeability of the core to which the coil is wrapped around, the greater the inductance.

Coil Length

The longer the coil's length, the lesser the inductance. The shorter the coil's length, the greater the inductance.

Types of Inductance

Inductance is classified into two types as:

Self Inductance

Mutual Inductance

Self Inductance

When there is a change in the current or magnetic flux of the coil, an electromotive force is induced. This phenomenon is termed Self Inductance. When the current starts flowing through the coil at any instant, it is found that, that the magnetic flux becomes directly proportional to the current passing through the circuit. The relation is given as:

$$\phi = L \times I$$

Where L is termed as the self-inductance of the coil or the coefficient of self-inductance, the self-inductance depends on the cross-sectional area, the permeability of the material, and the number of turns in the coil.

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The rate of change of magnetic flux in the coil is given as,

$$e = -\frac{d\phi}{dt} = -\frac{d(LI)}{dt}$$

$$e = -L \frac{dI}{dt}$$

Self Inductance Formula $L = N \frac{\phi}{I}$

Where,

L is the self inductance in Henries

N is the number of turns

Φ is the magnetic flux

I is the current in amperes

Mutual inductance

Consider two coils: P – coil (Primary coil) and S – coil (Secondary coil). A battery and a key are connected to the P-coil, whereas a galvanometer is connected across the S-coil. When there is a change in the current or magnetic flux linked with the two coils, an opposing electromotive force is produced across each coil, and this phenomenon is termed Mutual Inductance.

This phenomenon is given by the relation:

$$\phi = MI$$

Where M is termed as the mutual inductance of the two coils or the coefficient of the mutual inductance of the two coils.

The rate of change of magnetic flux in the coil is given as,

$$e = -\frac{d\phi}{dt} = -\frac{d(MI)}{dt}$$

$$e = -M \frac{dI}{dt}$$

Mutual Inductance Formula $M = \frac{\mu_0 \mu_r N_1 N_2 A}{l}$

Where,

μ_0 is the permeability of free space

μ_r is the relative permeability of the soft iron core

N is the number of turns in coil

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A is the cross-sectional area in m^2

l is the length of the coil in m

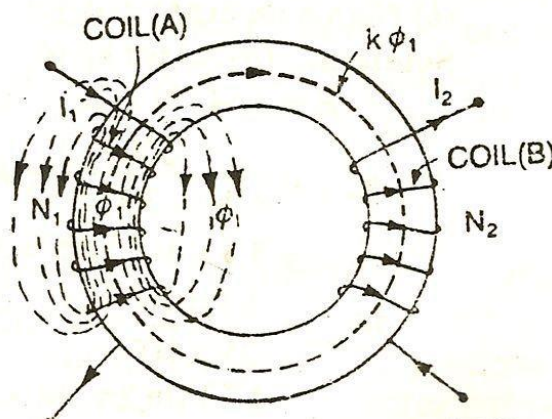
Difference between Self and Mutual Inductance

Self inductance	Mutual inductance
In self inductance, the change in the strength of current in the coil is opposed by the coil itself by inducing an e.m.f.	In mutual inductance out of the two coils one coil opposes change in the strength of the current flowing in the other coil.
The induced current opposes the growth of current in the coil when the main current in the coil increases.	The induced current developed in the neighboring coil opposes the growth of current in the coil when the main current in the coil increases.
The induced current opposes the decay of current in the coil when the main current in the coil decreases.	The induced current developed in the neighboring coil opposes the decay of the current in the coil when the main current in the coil decreases.

Coefficient of coupling.

The fraction of magnetic flux produced by the current in one coil that links with the other coil is called the coefficient of coupling between the two coils. It is denoted by (k) .

Two coils are taken coil A and coil B, when current flows through one coil it produces flux; the whole flux may not link with the other coil coupled, and this is because of leakage flux by a fraction (k) known as Coefficient Of Coupling.



$k=1$, when the flux produced by one coil, completely links with the other coil and is called magnetically tightly coupled.

$k=0$, when the flux produced by one coil, does not link at all with the other coil and thus the coils are said to be magnetically isolated.

Derivation

Consider two magnetic coils A and B.

When current I_1 flows through coil A.

$$L_1 = \frac{N_1 \phi_1}{I_1} \text{ and } M = \frac{N_2 \phi_{12}}{I_1} \dots \dots \dots (1) \text{ as } (\phi_{12} = k \phi_1)$$

Considering coil B in which current I_2 flows

$$L_2 = \frac{N_2 \phi_2}{I_2} \text{ and } M = \frac{N_1 \phi_{21}}{I_2} = \frac{N_1 k \phi_2}{I_2} \dots \dots \dots (2) \text{ as } (\phi_2 = k \phi_2)$$

Multiplying equation (1) and (2)

$$M \times M = \frac{N_2 k \phi_1}{I_1} \times \frac{N_1 k \phi_2}{I_2}$$

$$M^2 = k^2 \frac{N_1 \phi_1}{I_1} \times \frac{N_2 \phi_2}{I_2} = k^2 L_1 L_2$$

$$M = \sqrt{L_1 L_2} \dots \dots \dots (A)$$

The above equation (A) shows the relationship between mutual inductance and self-inductance between the two coils

DC Machines:

DC Generator: Construction and principle of operation

What is a DC Generator?

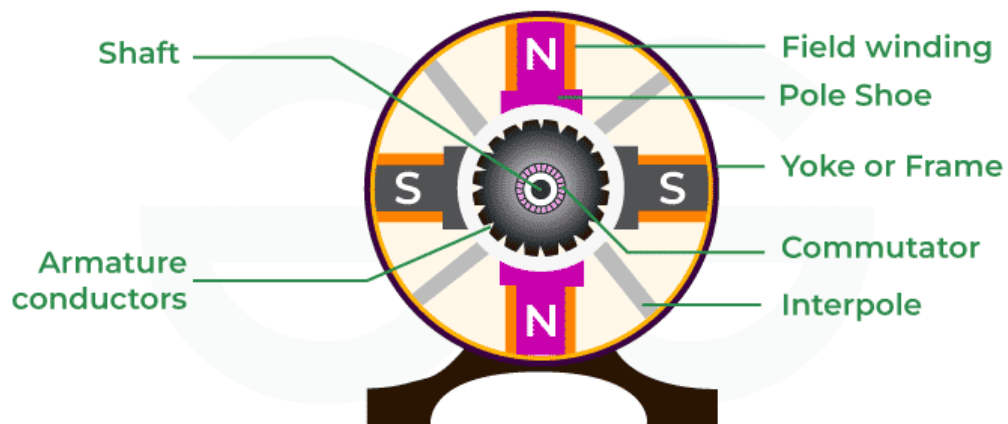
An electromechanical energy conversion device known as a DC generator uses electromagnetic principles to convert mechanical power into DC electrical power. According to the electromagnetic induction theory, an EMF is induced in a conductor when the magnetic flux connecting them changes. This is how a DC generator works. In a DC generator, there are two windings: the field and the armature.

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The EMF made in the armature Winding of a DC generator is rotating and is switched over completely to coordinate voltage by a commutator introduced on the generator's shaft. A DC generator's armature winding is on the rotor, while the field winding is on the stator.

Construction of a DC Generator

Below is a schematic representation of a DC generator:



Components of DC Generator

Components of DC Generator are given below:

Yoke

Magnetic Field System

Armature Core

Armature Winding

Commutator

DC Generator Commutator Function

Shaft

Pole Shoe

Poles

Bearings

Yoke

The outside frame of a DC generator is a hollow cylinder made of cast or rolled steel, known as a yoke. The yoke provides the following two objectives.

- It supports the field pole core and serves as a protective cover for the machine.
- It offers a channel for the magnetic flux produced by the field coil.

Magnetic Field System

A DC generator's magnetic field system serves as its stationary component. It generates the primary magnetic flux in the generator. It is made up of a large number of pole cores fastened to the yoke with field winding coiled around the pole core. The field arrangement of a DC generator features prominent poles, such as poles that thrust inwards, and each pole core has a post shoe with a curved surface. The shaft shoe has two objectives.

- It provides support for the field coils.
- It lowers the reluctance of a magnetic circuit by expanding its cross-sectional area.

To prevent eddy current loss, the pole cores are constructed of thin, insulated sheet steel laminations. The field coils are linked in series to create alternating north and south poles in the rotating direction as the current flows through them.

Armature Core

The armature core of a DC generator is positioned on a shaft and pivots between the field poles. It features slots on its exterior surface where the armature conductors are installed. The armature core is made up of soft iron laminations that are isolated from one another and securely attached together. In big machines, the laminations are keyed, but in tiny machines, they are keyed directly to the shaft. The laminated armature core helps to decrease eddy current losses.

Armature Winding

The insulated conductors are inserted into the slots of the armature core. The wires are properly connected. The linked arrangement of conductors is known as armature winding. There are two types of armature windings used: wave winding and lap winding.

Commutator

A commutator is a mechanical rectifier that converts the alternating magnetic field of the armature winding into a direct voltage that runs through the load terminals.

The commutator is constructed up of wedge-shaped copper segments that are isolated from one another and the shaft by mica sheets. Each commutator segment is connected to one of the armature coil's ends.

DC Generator Commutator Function

The basic purpose of the commutator is to convert DC to AC. It functions as a reversing switch, and its purpose in the generator is explained below.

The electromagnetic field produced within the generator's armature coil alternates, leading to an alternating current flowing within the armature coil. This current may be reversed through the commutator at the precise moment that the armature coil crosses the magnetic unbiased axis. As a result, the load develops a DC or unidirectional current.

The commutator ensures that the current flowing from the generator continues in the same direction indefinitely. By moving along the commutator, the brushes create high-quality electrical connections between the generator and the load.

Shaft

The shaft is a key component of a DC machine because it produces torque, which causes rotation. It is constructed of mild steel and has the highest breaking strength. The shaft is a component of a DC generator that influences the generator's ability to transport mechanical energy. The shaft is keyed to the commutator, cooling fan, armature center, and other spinning components.

Pole Shoe

A pole shoe is a plate composed of iron or steel that spreads magnetic flux and prevents field coils from falling.

Poles

Poles help maintain the field windings in excellent shape. These windings are generally coiled on poles and linked in a certain order to the armature windings. As a result, the posts attach the winding procedure to the yoke using screws.

Bearings

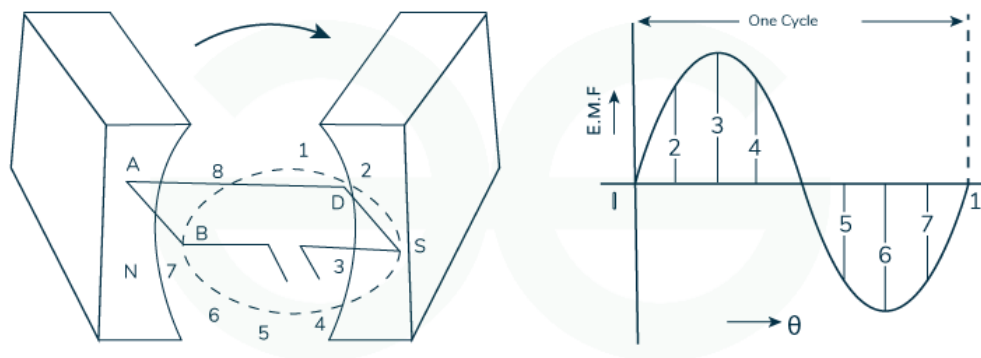
Bearings are used in a system to ensure that the various parts of a DC machine move smoothly. The friction between the machine's spinning and stationary elements is reduced with the aid of course. As a result, the system's components require less

frequent greasing and will last longer. Roller bearings and ball bearings are the two most popular types of bearings used in dc generators.

Workings of a DC Generator

Consider a single loop DC generator (as seen in the diagram), in which a single turn loop 'ABCD' rotates clockwise in a uniform magnetic field at constant speed. As the loop rotates, the magnetic flux between the coil sides 'AB' and 'CD' changes continually. This change in flux linkage causes an EMF to be induced in both coil sides, and the induced EMF on one coil side is added to the induced EMF on the other.

Induced emf expression



The EMF produced by a DC generator may be explained as follows:

1. When the loop is at position 1, no EMF is produced since the coil sides move in parallel with the magnetic flux.
2. When the loop is in position 2, the loop sides move in the direction of the attracting motion, generating a tiny EMF.
3. When the loop is at place 3, the coil sides move at the right point to the attractive transition, resulting in the highest EMF.
4. When the loop is in position 4, the coil sides cut the attractive transition at a point, resulting in a lower EMF in the coil sides.
5. When the loop is at place 5, there is no motion coupling with the coil side, and the movement is aligned with the attractive transition. Consequently, no EMF is produced in the coil.
6. At position 6, the coil sides move beneath an inverse extremity post, switching the extremity of the generated EMF. The most intense EMF will be produced toward this route at position-7, and zero at position-1. This cycle repeats for each coil turn.

It is evident that the produced EMF is a spinning one. It is because any coil side (for example, AB) has EMF in one direction when impacted by an N-pole and in the

opposite direction when affected by an S-pole. As a result, when a load is attached to the generator's terminals, alternating current flows through it. Currently, using a commutator, the alternating emf created in the loop may be converted into direct voltage.

Derivation of Induced EMF in a DC Machine

When the armature of a DC machine rotates in a magnetic field, an EMF is induced in its conductors according to **Faraday's law of electromagnetic induction**.

Step 1: EMF Induced in One Conductor

Faraday's Law:

$$e = \frac{d\Phi}{dt}$$

A conductor rotates and passes through one pole flux (ϕ).

In **one revolution**, the flux cut by one conductor:

$$\text{Flux cut} = P\Phi$$

where

P = number of poles

ϕ = flux per pole (Webers)

Step 2: Time Taken for One Revolution

If the armature rotates at:

N revolutions per minute (rpm)

Then time taken for **one revolution**:

$$t = \frac{60}{N} \text{ seconds}$$

Step 3: EMF per Conductor

$$e_c = \frac{\text{Flux cut}}{\text{Time}} = \frac{P\Phi}{\frac{60}{N}} = \frac{P\Phi N}{60}$$

Step 4: Total EMF in Armature

Let

Z = total number of armature conductors

These conductors are divided into (A) parallel paths.

Therefore, number of conductors per parallel path:

$$\frac{Z}{A}$$

Since conductors in each path are in series, EMF in each path:

$$E = e_c \times \frac{Z}{A}$$

Substituting $e_c = \frac{P\Phi N}{60}$:

$$E = \frac{P\Phi N}{60} \times \frac{Z}{A}$$

Final Expression

$$E = \frac{P\Phi ZN}{60A}$$

Where

(E) Induced EMF in armature (Volts)

(P) Number of poles

(Φ) Flux per pole (Weber)

(Z) Total number of conductors

(N) Armature speed (RPM)

(A) Number of parallel paths

Lap winding $\rightarrow$ (A = P) Wave winding $\rightarrow$ (A = 2)

Classification of generators with relation between induced emf and terminal voltage.

There are three techniques of excitation, and hence three primary types of DC generators:

Permanent Magnet DC Generators - Field coils energized by permanent magnets

Separately stimulated DC Generators: Field coils stimulated by an external source.

Self-excited DC generators have field coils that are stimulated by the generator itself.

These generators are further classified based on their field coil location. They are:

Series Generators

Shunt Generators

Compound Generators

DC Motor:

Back emf and its significance

Back EMF (Electromotive Force), also known as counter EMF, is a voltage induced in the armature of a DC motor as it rotates, opposing the applied voltage. This induced voltage arises from the motor's conductors moving through the magnetic field, and its polarity is such that it resists the change in current that created it.

Significance of Back EMF:

Speed Regulation:

Back EMF is directly proportional to the motor's speed. As the motor speeds up, the back EMF increases, reducing the net voltage across the armature and thus the armature current. This limits the motor's speed and prevents it from over-speeding. Conversely, when the motor slows down due to increased load, the back EMF decreases, allowing more current to flow and provide the necessary torque to maintain the speed.

Current Limiting:

When a DC motor is first energized, the armature is not rotating, and the back EMF is minimal. This allows a large current to flow initially, which can damage the motor. However, as the motor starts rotating and generating back EMF, it limits the current flow to a safe level.

Torque Generation:

The electrical work done by the applied voltage is converted into mechanical energy (torque) and back EMF. The back EMF opposes the applied voltage, and the motor converts this opposing voltage into torque.

Energy Conservation:

Back EMF is a manifestation of energy conservation in a DC motor. The electrical energy supplied to the motor is converted into both mechanical energy (shaft work) and back EMF, which is a form of stored electrical energy.

Torque equation

When a DC machine is loaded either as a motor or as a generator, the rotor conductors carry current. These conductors lie in the magnetic field of the air gap.

Thus, each conductor experiences a force. The conductors lie near the surface of the rotor at a common radius from its centre. Hence, a torque is produced around the circumference of the rotor, and the rotor starts rotating.

When the machine operates as a generator at a constant speed, this torque is equal and opposite to that provided by the prime mover.

When the machine is operating as a motor, the torque is transferred to the shaft of the rotor and drives the mechanical load. The expression is the same for the generator and motor.

When the current-carrying conductor is placed in the magnetic field, a force is exerted on it which exerts turning moment or torque $F \times r$. This torque is produced due to the electromagnetic effect, hence is called Electromagnetic torque.

The torque which is produced in the armature is not fully used at the shaft for doing the useful work. Some part of it gets lost due to mechanical losses. The torque which is used for doing useful work is known as the shaft torque.

Since,

$$V = E_b + I_a R_a \dots\dots\dots (1)$$

Multiplying the equation (1) by I_a we get

$$VI_a = E_b I_a + I_a^2 R_a \dots\dots\dots (2)$$

Where,

VI_a is the electrical power input to the armature.

$I_a^2 R_a$ is the copper loss in the armature.

We know that,

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Total electrical power supplied to the armature = Mechanical power developed by the armature + losses due to armature resistance

Now, the mechanical power developed by the armature is P_m ,

$$P_m = E_b I_a \dots \dots (3)$$

Also, the mechanical power that rotates the armature can be given regarding torque T and speed n .

$$P_m = \omega T = 2\pi nT \dots \dots (4)$$

Where n is in revolution per seconds (rps) and T is in Newton-Meter.

Hence,

$$2\pi nT = E_b I_a \quad \text{or}$$

$$T = \frac{E_b I_a}{2\pi n}$$

But,

$$E_b = \frac{\phi ZNP}{60 A}$$

Where N is the speed in revolution per minute (rpm) and

$$n = \frac{N}{60}$$

where n is the speed in (rps).

Therefore,

$$E_b = \frac{\phi ZnP}{A}$$

So, the torque equation is given as:

$$T = \frac{\phi ZP}{2\pi A} \cdot I_a$$

For a particular DC Motor, the number of poles (P) and the number of conductors per parallel path (Z/A) are constant.

$$T = K\phi I_a$$

Where

$$K = \frac{ZP}{2\pi A} \quad \text{or}$$

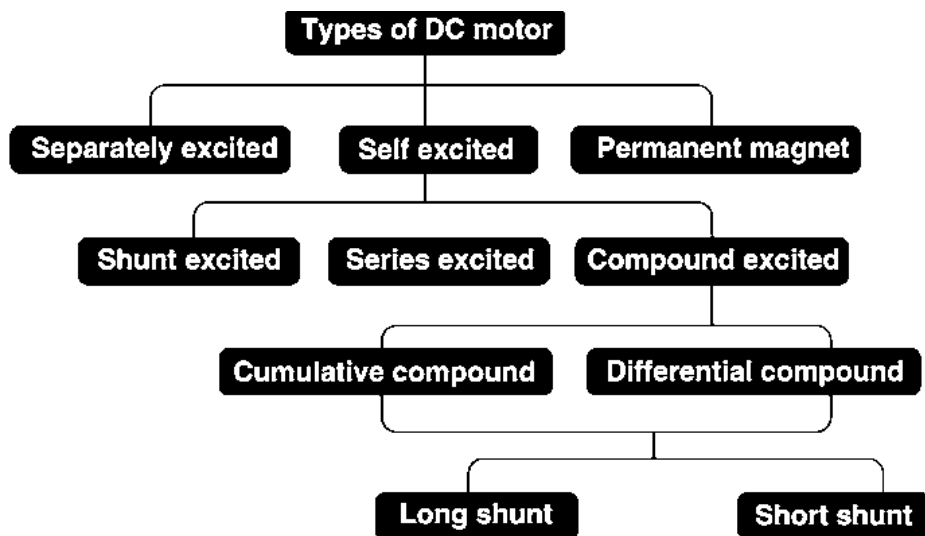
$$T \propto \phi I_a \dots \dots (5)$$

Thus, from the above equation (5) it is clear that the torque produced in the armature is directly proportional to the flux per pole and the armature current.

Moreover, the direction of electromagnetic torque developed in the armature depends upon the current in armature conductors. If either of the two is reversed the direction of torque produced is reversed and hence the direction of rotation. But when both are reversed, and direction of torque does not change.

Classification

Based on the type of construction and electrical connection, DC motors can be categorised into four major types. A brief chart (based on the type of construction and electrical connection) is given below:



Types of DC motor when listed sequentially:

Permanent Magnet DC Motor

Shunt Motor

Series Motor

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Compound Motor is again classified into

Short Shunt Motor

Long Shunt Motor

Differentially Compound

Cumulative Compound

Permanent Magnet DC Motor

In these types of DC motor, the permanent magnet is used to create a magnetic field. In this, no input current is consumed for excitation. These are used in automobile starters, wipers, air conditioners, etc

Shunt DC Motor

Here, the field is connected with the armature windings in parallel or also known as a shunt. The shunt field can be separately excited from the armature windings and that is the reason it can be used for greater speed regulation and can also offer very simplified reverse control.

Series DC Motor

Here, a large wire carrying the full armature current winds the field with few turns. This kind of motor generates a large amount of starting torque but the speed cannot be regulated here. If they are run with no load then it might face damage. These are not the ideal option for variable speed applications.

Compound DC Motor

These have a shunt field which is separately excited. They have a good starting torque but might face problems in variable speed applications.

Short Shunt DC Motor

Here, the shunt field winding is only connected with the armature winding and that too in parallel. The field coil, which is in series, is entirely exposed to current before being split up into the armature.

Long Shunt DC Motor

Here, the shunt field winding is connected in parallel with the both series field coil and armature which are again connected with each other in series.

Differentially Compound DC Motor

In this type of compound wound DC motor, the flux produced due to the shunt field windings reduces the effect of the main series windings.

Cumulative Compound DC Motor

Here, the flux produced by the shunt field windings enhances the effect of the main field flux which is produced by series winding.

Characteristics

The performance of a DC motor is given by the relation among the armature current, torque and speed. These relations are given graphically in the form of curves, which are called as characteristics of DC motors. These characteristics show the behaviour of the DC motor under different load conditions.

Following are the three important characteristics of a DC motor

Torque and Armature Current Characteristics

It is the graph plotted between the armature torque (τ_a) and the armature current (I_a) of a DC motor. It is also known as electrical characteristics of the DC motor.

Speed and Armature Current Characteristics

It is the graph plotted between the speed (N) and the armature current (I_a) of a DC motor. This characteristic curve is mainly used for selecting a motor for a particular application.

Speed and Torque Characteristics

The graph plotted between the speed (N) and the armature torque (τ_a) for a DC motor is known as the speed-torque characteristics. It is also known as mechanical characteristics of DC motor.

Applications.

DC Motor Types and Their Applications

Shunt Wound Motors:

Characteristics: Maintain a nearly constant speed, even when the load changes.

Applications: Used where a stable speed is required, such as in centrifugal pumps, lathe machines, and drilling machines.

Series Wound Motors:

Characteristics: Provide very high starting torque but their speed varies significantly with the load.

Applications: Ideal for high-inertia loads and applications requiring high starting power, such as electric trains, cranes, hoists, and conveyors.

Compound Wound Motors:

Characteristics: Combine characteristics of both shunt and series motors, offering a balance of stable speed and high starting torque.

Applications: Suitable for loads that fluctuate, such as elevators, pressurized blowers, and shears.

Permanent Magnet DC Motors:

Characteristics: Highly efficient, compact, and low-maintenance.

Applications: Commonly used in small applications like fans, pumps, and servo motors.

Brushless DC Motors:

Characteristics: Known for their high efficiency, long lifespan, and low maintenance.

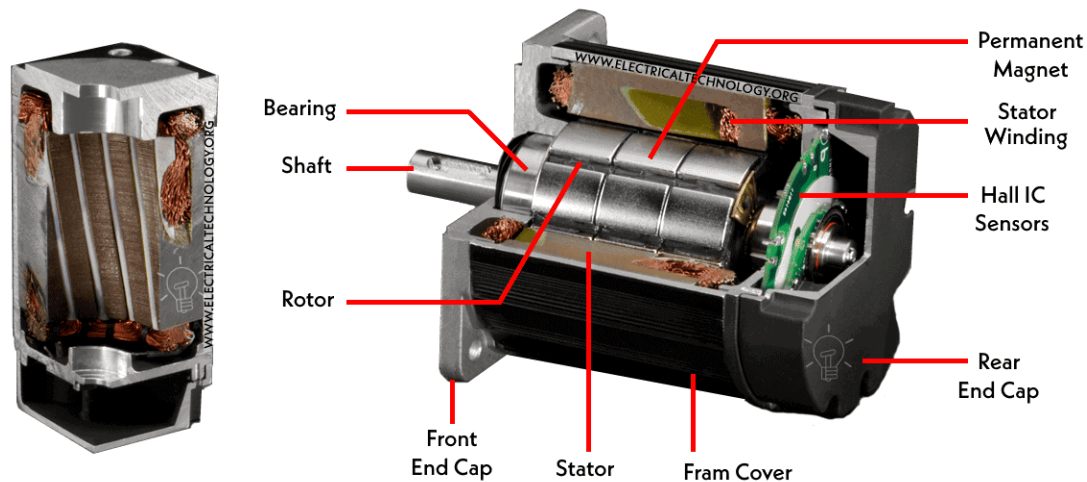
Applications: Used in applications requiring high reliability and efficiency, such as robotics, HVAC systems, and high-precision industrial equipment.

EV Motors:

BLDC Motor

A brushless DC motor (known as BLDC) is a permanent magnet synchronous electric motor which is driven by direct current (DC) electricity and it accomplishes electronically controlled commutation system (commutation is the process of producing rotational torque in the motor by changing phase currents through it at

appropriate times) instead of a mechanical commutation system. BLDC motors are also referred as trapezoidal permanent magnet motors.



Unlike conventional brushed type DC motor, wherein the brushes make the mechanical contact with commutator on the rotor so as to form an electric path between a DC electric source and rotor armature windings, BLDC motor employs electrical commutation with permanent magnet rotor and a stator with a sequence of coils. In this motor, permanent magnet (or field poles) rotates and current carrying conductors are fixed.

The armature coils are switched electronically by transistors or silicon-controlled rectifiers at the correct rotor position in such a way that armature field is in space quadrature with the rotor field poles. Hence the force acting on the rotor causes it to rotate. Hall sensors or rotary encoders are most commonly used to sense the position of the rotor and are positioned around the stator. The rotor position feedback from the sensor helps to determine when to switch the armature current.

This electronic commutation arrangement eliminates the commutator arrangement and brushes in a DC motor and hence more reliable and less noisy operation is achieved. Due to the absence of brushes BLDC motors are capable to run at high speeds. The efficiency of BLDC motors is typically 85 to 90 percent, whereas as brushed type DC motors are 75 to 80 percent efficient. There are wide varieties of BLDC motors available ranging from small power range to fractional horsepower, integral horsepower and large power ranges.

Construction of BLDC Motor

BLDC motors can be constructed in different physical configurations. Depending on the stator windings, these can be configured as single-phase, two-phase, or three-phase motors. However, three-phase BLDC motors with permanent magnet rotor are most commonly used.

The construction of this motor has many similarities of three phase induction motor as well as conventional DC motor. This motor has stator and rotor parts as like all other motors.

Stator

Stator of a BLDC motor made up of stacked steel laminations to carry the windings. These windings are placed in slots which are axially cut along the inner periphery of the stator. These windings can be arranged in either star or delta. However, most BLDC motors have three phase star connected stator.

Each winding is constructed with numerous interconnected coils, where one or more coils are placed in each slot. In order to form an even number of poles, each of these windings is distributed over the stator periphery.

The stator must be chosen with the correct rating of the voltage depending on the power supply capability. For robotics, automotive and small actuating applications, 48 V or less voltage BLDC motors are preferred. For industrial applications and automation systems, 100 V or higher rating motors are used.

Rotor

BLDC motor incorporates a permanent magnet in the rotor. The number of poles in the rotor can vary from 2 to 8 pole pairs with alternate south and north poles depending on the application requirement. In order to achieve maximum torque in the motor, the flux density of the material should be high. A proper magnetic material for the rotor is needed to produce required magnetic field density.

Ferrite magnets are inexpensive; however they have a low flux density for a given volume. Rare earth alloy magnets are commonly used for new designs. Some of these alloys are Samarium Cobalt (SmCo), Neodymium (Nd), and Ferrite and Boron (NdFeB). The rotor can be constructed with different core configurations such as the circular core with permanent magnet on the periphery, circular core with rectangular magnets, etc.

Hall Sensors

Hall sensor provides the information to synchronize stator armature excitation with rotor position. Since the commutation of BLDC motor is controlled electronically, the stator windings should be energized in sequence in order to rotate the motor. Before energizing a particular stator winding, acknowledgment of rotor position is necessary. So the Hall Effect sensor embedded in stator senses the rotor position.

Most BLDC motors incorporate three Hall sensors which are embedded into the stator. Each sensor generates Low and High signals whenever the rotor poles pass near to it. The exact commutation sequence to the stator winding can be determined based on the combination of these three sensor's response.

Working of Brushless DC Motor

BLDC motor works on the principle similar to that of a conventional DC motor, i.e., the Lorentz force law which states that whenever a current carrying conductor placed in a magnetic field it experiences a force. As a consequence of reaction force, the magnet will experience an equal and opposite force. In case BLDC motor, the current carrying conductor is stationary while the permanent magnet moves.

When the stator coils are electrically switched by a supply source, it becomes electromagnet and starts producing the uniform field in the air gap. Though the source of supply is DC, switching makes to generate an AC voltage waveform with trapezoidal shape. Due to the force of interaction between electromagnet stator and permanent magnet rotor, the rotor continues to rotate.

Consider the figure below in which motor stator is excited based on different switching states. With the switching of windings as High and Low signals, corresponding winding energized as North and South poles. The permanent magnet rotor with North and South poles align with stator poles causing motor to rotate.

Observe that motor produces torque because of the development of attraction forces (when North-South or South-North alignment) and repulsion forces (when North-North or South-South alignment). By this way motor moves in a clockwise direction.

Here, one might get a question that how we know which stator coil should be energized and when to do. This is because; the motor continuous rotation depends on the switching sequence around the coils. As discussed above that Hall sensors give shaft position feedback to the electronic controller unit.

Based on this signal from sensor, the controller decides particular coils to energize. Hall-effect sensors generate Low and High level signals whenever rotor poles pass near to it. These signals determine the position of the shaft.

Permanent Magnet Synchronous Motor

The permanent magnet synchronous motors are one of the types of AC synchronous motors, where the field is excited by permanent magnets that generate sinusoidal back EMF. It contains a rotor and stator same as that of an induction motor, but a permanent magnet is used as a rotor to create a magnetic field. Hence there is no need to wound field winding on the rotor. It is also known as a 3-phase brushless permanent sine wave motor. The permanent magnet synchronous motor diagram is shown below.

Working Principle

The permanent magnet synchronous motor working principle is similar to the synchronous motor. It depends on the rotating magnetic field that generates electromotive force at synchronous speed. When the stator winding is energized by giving the 3-phase supply, a rotating magnetic field is created in between the air gaps.

This produces the torque when the rotor field poles hold the rotating magnetic field at synchronous speed and the rotor rotates continuously. As these motors are not self-starting motors, it is necessary to provide a variable frequency power supply.

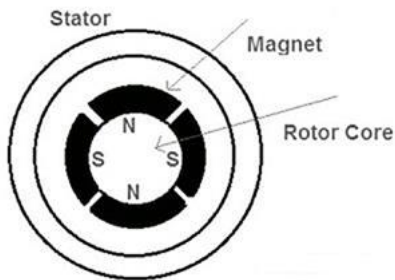
Construction

The permanent magnet synchronous motor construction is similar to the basic synchronous motor, but the only difference is with the rotor. The rotor doesn't have any field winding, but the permanent magnets are used to create field poles. The permanent magnets used in the PMSM are made up of samarium-cobalt and medium, iron, and boron because of their higher permeability.

The most widely used permanent magnet is neodymium-boron-iron because of its effective cost and ease of availability. In this type, the permanent magnets are mounted on the rotor. Based on the mounting of the permanent magnet on the rotor, the construction of a permanent magnet synchronous motor is divided into two types. They are,

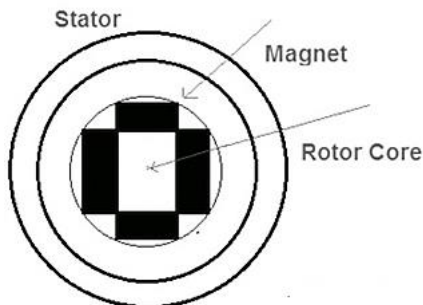
Surface-mounted PMSM

In this construction, the magnet is mounted on the surface of the rotor. It is suited for high-speed applications, as it is not robust. It provides a uniform air gap because the permeability of the permanent magnet and the air gap is the same. No reluctance torque, high dynamic performance, and suitable for high-speed devices like robotics and tool drives.



Buried PMSM or Interior PMSM

In this type of construction, the permanent magnet is embedded into the rotor as shown in the figure below. It is suitable for high-speed applications and gets robustness. Reluctance torque is due to the saliency of the motor.



Advantages

The advantages of permanent magnet synchronous motor include,

- provides higher efficiency at high speeds
- available in small sizes at different packages
- maintenance and installation is very easy than an induction motor
- capable of maintaining full torque at low speeds.
- high efficiency and reliability
- gives smooth torque and dynamic performance

Disadvantages

The disadvantages of permanent magnet synchronous motors are,

Unit-III

- These type of motors are very expensive when compared to induction motors
- Somehow difficult to start-up because they are not self-starting motors.

Applications

The permanent magnet synchronous motors applications are,

- Air conditioners
- Refrigerators
- AC compressors
- Washing machines, which are direct-drive
- Automotive electrical power steering
- Machine tools
- Large power systems to improve leading, and lagging power factor
- Control of traction
- Data storage units.
- Servo drives
- Industrial applications like robotics, aerospace, and many more.

Permanent Magnet Synchronous Motor vs BLDC

The differences between permanent magnet synchronous motor (PMSM) and BLDC ([Brushless DC motors](#)) include the following.

Permanent Magnet Synchronous Motor	BLDC Motor
These are brushless AC synchronous motors	These are brushless DC motors
Torque ripples are absent	Torque ripples are present
Performance efficiency is high	Performance efficiency is low
More efficient	Less efficient
Used in industrial applications, automobiles, servo motors, robotics, train drives, etc	Used in electronic steering power systems, HVAC systems, hybrid train drives (electrical), etc
Produces low noise	Produces high noise.